

Question:

A silicious rock contains the mineral ZnS. To analyze for Zn, a sample of the rock is pulverized and treated with HCl to dissolve the ZnS (silicious matter is insoluable). Zinc is precipitated from solution by the addition of potassium ferrocyanide $K_4Fe(CN)_6$. After filtering, the precipitate is dried and weighed. The reactions which occur are $ZnS + 2HCl \rightarrow ZnCl_2 + H_2S$ $2ZnCl_2 + K_4Fe(CN)_6 \rightarrow Zn_2Fe(CN)_6 + 4KCl$ If a 2 gram sample of rock yields 0.969 gram of $Zn_2Fe(CN)_6$, what is the percentage of Zn in the sample? Atomic weight Zn = 65.4, molecular weight $Zn_2Fe(CN)_6 = 342.6$.

Answer Choices:

- A. 10 % Zn
- B. 22 % Zn
- C. 20 % Zn
- D. 19 % Zn
- E. 15 % Zn
- F. 18 % Zn
- G. 16 % Zn
- H. 14 % Zn
- I. 12 % Zn
- J. 25 % Zn

Reasoning + Answer:

<think>

- Zn content in $Zn_2Fe(CN)_6$ can be calculated by taking ratio of the weight of Zn to the weight of $Zn_2Fe(CN)_6$.
- Weight of Zn in $Zn_2Fe(CN)_6 = 2 * 65.4 g = 130.8 g$.
- Weight of $Zn_2Fe(CN)_6 = 342.6 g$.
- The ratio of Zn's weight to $Zn_2Fe(CN)_6$'s weight is $(130.8 / 342.6) = 0.3815$ approx.
- Therefore, Zn from 2 g rock $= 0.969 g Zn_2Fe(CN)_6 * 0.3815 = 0.37 g Zn$.
- The % of Zn in sample $= (0.37 g Zn / 2 g rock) * 100 = 18.5 \% \hat{=} 18 \%$ approx.

</think>

<answer>

F. 18 % Zn

</answer>